

12a) Use a adaptive Network to learn ANDNOT function with bipolar inputs and targets perform 2 passes of training.

b) state the concept of delta rule used in adapting linear neurons.

Solutions

x_1	x_2	x_3	t
1	1	1	-1
1	-1	1	1
-1	1	1	-1
-1	-1	1	-1

$$w_1 = w_2 = b = 0.2$$

$$x_1 = 1 \quad x_2 = 1 \quad t = -1$$

$$y_{in} = b + w_1 x_1 + w_2 x_2$$

$$= 0.2 + 1 \times 0.2 + 1 \times 0.2$$

$$= 0.6$$

$$t - y_{in} = -1.6$$

$$w_1(\text{new}) = w_1(\text{old}) + \alpha (t - y_{in}) x_1$$

$$= 0.2 + 0.2 \times (-1.6) = -0.12$$

$$w_2(\text{new}) = w_2(\text{old}) + \alpha (t - y_{in}) x_2$$

$$= 0.2 + 0.2 \times (-1.6) \times 1 = -0.12$$

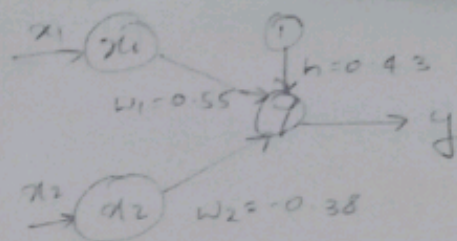
$$b(\text{new}) = b(\text{old}) + \alpha (t - y_{in})$$

$$= -0.12$$

$$E = (-1.6)^2 = 2.56$$

Inputs						Weights			Error (t - y _{in}) ²
x_1	x_2	x_3	t	y_{in}	(t - y _{in})	w_1 (0.2)	w_2 (0.2)	b (0.2)	
1	1	1	-1	0.6	-1.6	-0.12	-0.12	-0.12	2.56
1	-1	1	1	-0.12	1.12	0.10	-0.34	-0.10	1.25
-1	1	1	-1	-0.34	-0.66	0.24	-0.46	-0.03	0.43
-1	-1	1	-1	0.21	-1.2	0.46	-0.23	-0.27	1.47
1	1	1	-1	-0.02	-0.98	0.28	-0.43	-0.46	0.95
1	-1	1	1	0.25	0.76	0.43	-0.58	-0.31	0.57
-1	1	1	-1	-0.31	-0.33	0.37	-0.51	-0.25	0.46
-1	1	1	-1	-0.11	-0.90	0.55	-0.38	-0.43	0.8

Epoch	MSE
I	5.71
II	2.43



b) Concepts of Delta rule in Adaline

The delta rule (window-rhoff rule) is used in Adaptive Linear neurons (Adaline)

Key Concepts:

1. Based on gradient descent
 - Minimize Mean Squared Error (MSE)
2. Uses Linear Activation
 - Learning occurs before applying threshold
3. Error Calculation
 - $$\text{Error} = T - y_{in}$$
4. Weight Adjustment
 - Weights updated proportional to error and input
5. Objective
 - Reduce total squared error over training set
6. Continuous Learning
 - Provide smoother convergence compared to perceptron learning rule.